

INFOVIS REDESIGN

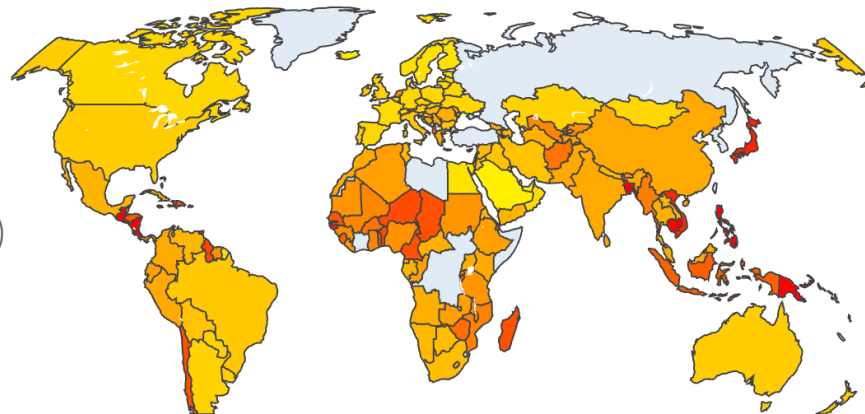
This project explores the redesign of a static world map visualization showing the Cooling Degree Days (CDD) from the World Risk Index dataset. The original design lacked crucial metadata, emotional engagement, and accessibility features. This redesign integrates perceptual theory, affective visualization research, and interactive tools to create a more insightful and inclusive visualization.

This project was guided by several theories:

- Tufte (1983)
- Munzner (2014)
- Fekete & Plaisant (1999)
- Buja et al. (1996)
- FAIR Principle
- Ziemkiewicz et al. (2021)

Tools Used

- Python: pandas, pycountry, Plotly



Workflow Pipeline

1. Data Processing: Cleaned and prepared CDD indicators using pandas
2. Country Mapping: Matched country names to ISO codes via pycountry
3. Visualization: Developed choropleth maps in Plotly for interactivity
4. User Options: Enabled dropdown filters (exposure, vulnerability, etc.)
5. Accessibility: Color-contrast checked; added tooltips and labels
6. Publication: Deployed code and dataset via GitHub for open access

Design Objectives

- Improve readability with clear labels and legends
- Apply emotional design principles to raise awareness
- Add interactivity for user-driven exploration
- Ensure metadata transparency (source, timeframe, indicators)
- Follow accessibility and perceptual optimization standards

LLM Result

Clarity & Readability

Strengths: Clear gradient to represent CDD increase

Weaknesses: No legend, no country labels, no timeframe

Effectiveness in Communicating Insights

Strengths: Global perspective, highlights high-risk areas

Weaknesses: Lacks specific data points, hard to compare trends

Aesthetic Appeal & Design

Strengths: Simple and visually engaging

Weaknesses: Color accessibility issues, missing interactive elements

Potential Improvements

- Add legend & numerical scale for precise interpretation
- Include country labels & annotations for key regions
- Provide historical comparisons to show trends
- Introduce interactive features for better user engagement